



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Data Collection
- Data Wrangling
- EDA with Data Visualization
- EDA with SQL
- Building an Interactive Map with Folium
- Building a Dashboard with Plotly Dash
- Predictive Analysis

Summary of all results

- EDA
- Interactive Analytics
- Predictive Analysis

Introduction

- **Project background and context**

SpaceX advertises Falcon 9 rocket launches with a cost of 62 million dollars while other providers can spend 165 million dollars each. The reason behind the savings of SpaceX is because they can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.

- **Problems you want to find answers**

If SpaceX can reuse the first stage, predicting if the first stage will land successfully is crucial to come up that indeed SpaceX can incur lesser cost for Falcon 9 rocket launches compared to other providers.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - SpaceX Rest API
 - Web Scrapping from Wikipedia
- Perform data wrangling
 - One Hot Encoding data fields for Machine Learning and data cleaning of null values and irrelevant columns.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Logistic Regression, KNN, SVM, DT models are built and evaluated for the best method.

Data Collection

- Describe how data sets were collected.

- SpaceX Rest API

SpaceX launch data was gathered from the SpaceX Rest API. The SpaceX REST API endpoints starts with `api.spacexdata.com/v4/`. The data include launches, rocket used, payload delivered, launch specs, landing specifications, and landing outcome.

URL used: `https://api.spacexdata.com/v4/launches/past`

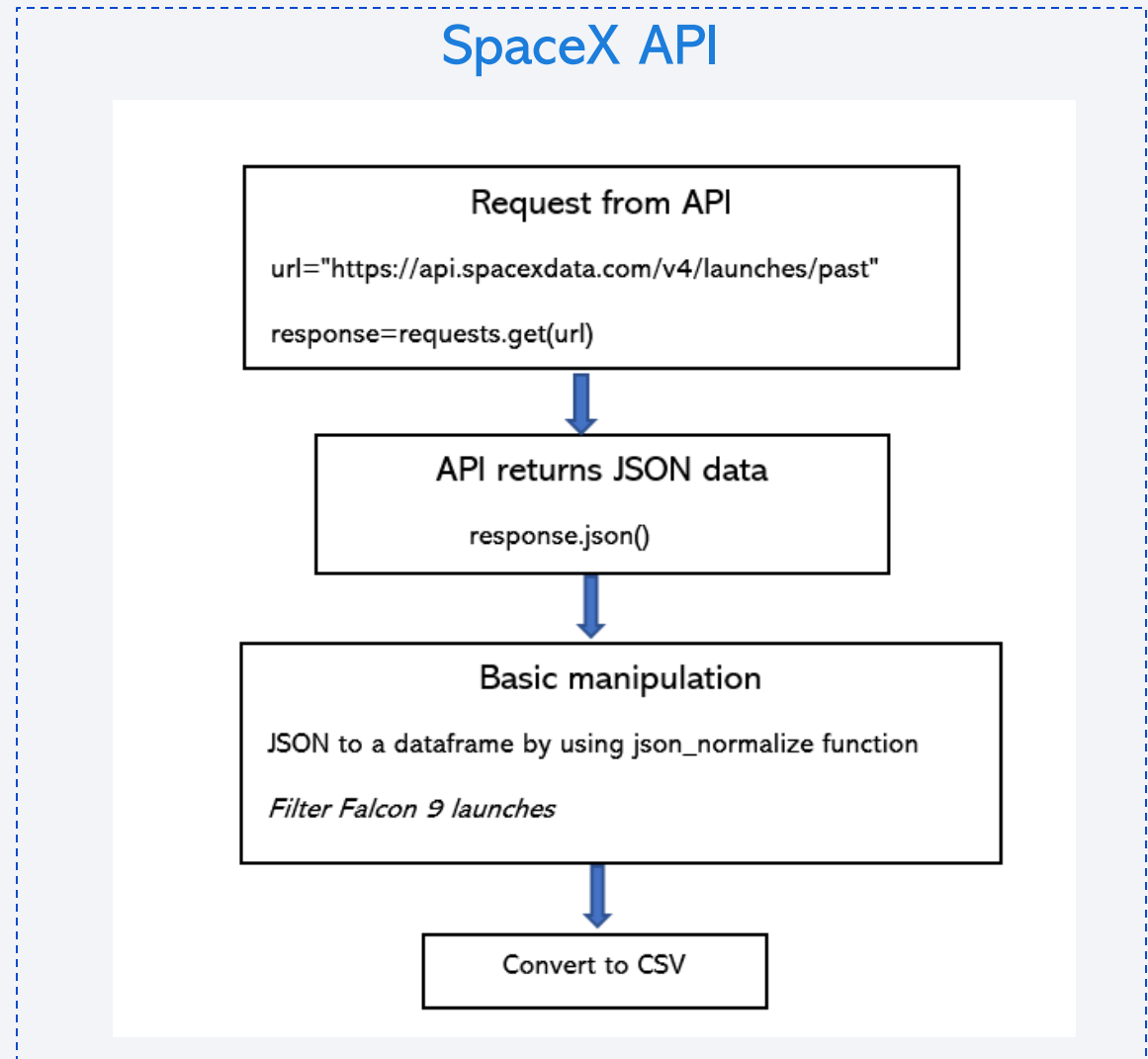
- Web Scrapping from Wikipedia

Additional data was scrapped from Wikipedia using BeautifulSoup to supplement the launch data.

- You need to present your data collection process use key phrases and flowcharts

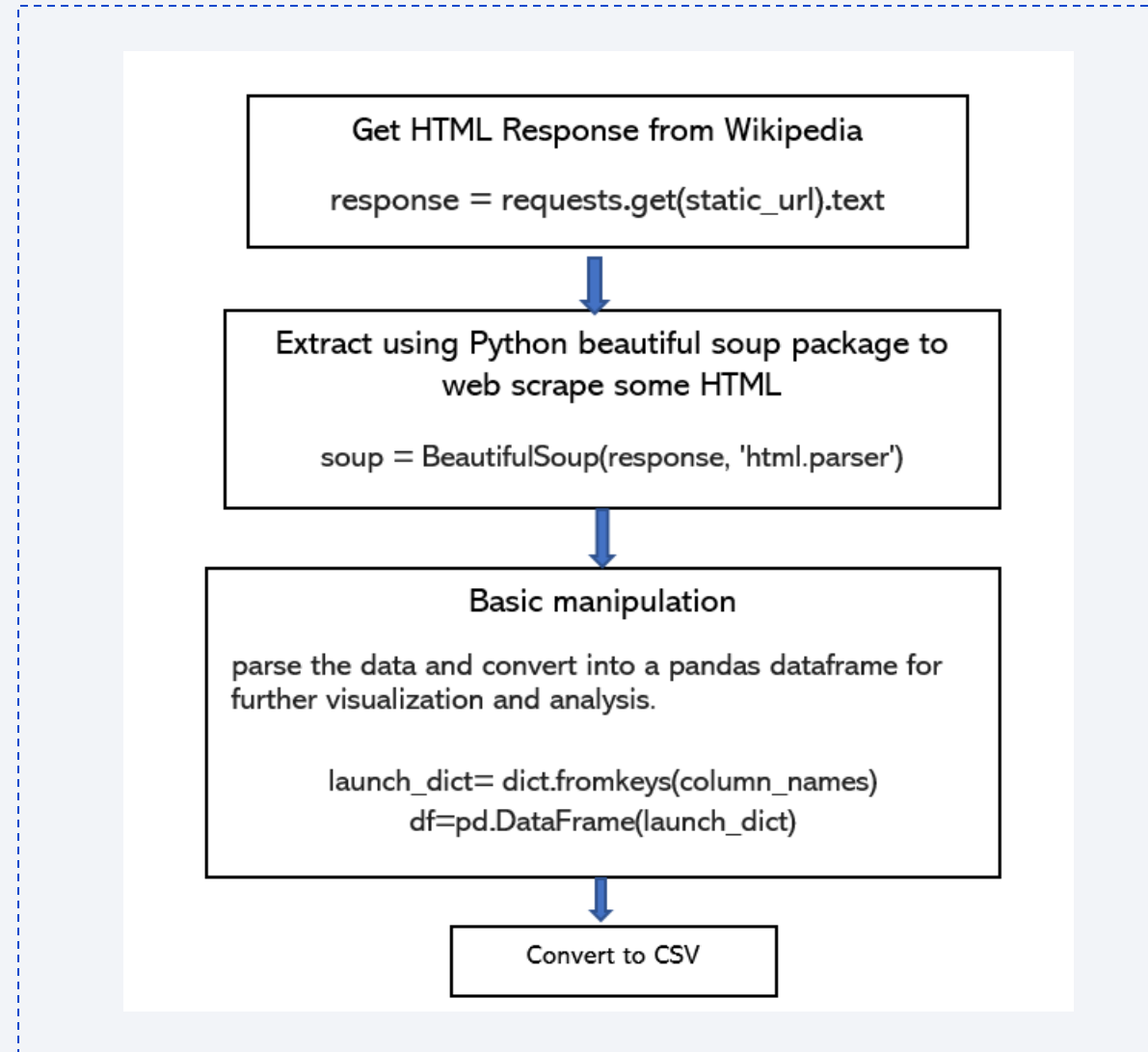
Data Collection – SpaceX API

- Get request is performed using the request library to get the launch data from the API. The result can be viewed by calling the .json() method. To convert the JSON to a dataframe, json_normalize function is used. This function allowed us to “normalize” the structured json data into a flat table.
- GitHub URL of the completed SpaceX API calls notebook:
<https://github.com/granaff/gafftestrepo/blob/1757cc84a14677a83fb97cc275c9642912951838/Data%20Collection%20API.ipynb>



Data Collection - Scraping

- Python BeautifulSoup package is used to web scrape some HTML tables from Wikipedia that contain valuable Falcon 9 launch records. Then, the data are parsed and convert them into Pandas dataframe for further visualization and analysis.
- Add the GitHub URL of the completed web scraping notebook:
<https://github.com/granaff/gafftestrepo/blob/1757cc84a14677a83fb97cc275c9642912951838/Data%20Collection%20with%20Web%20Scraping.ipynb>

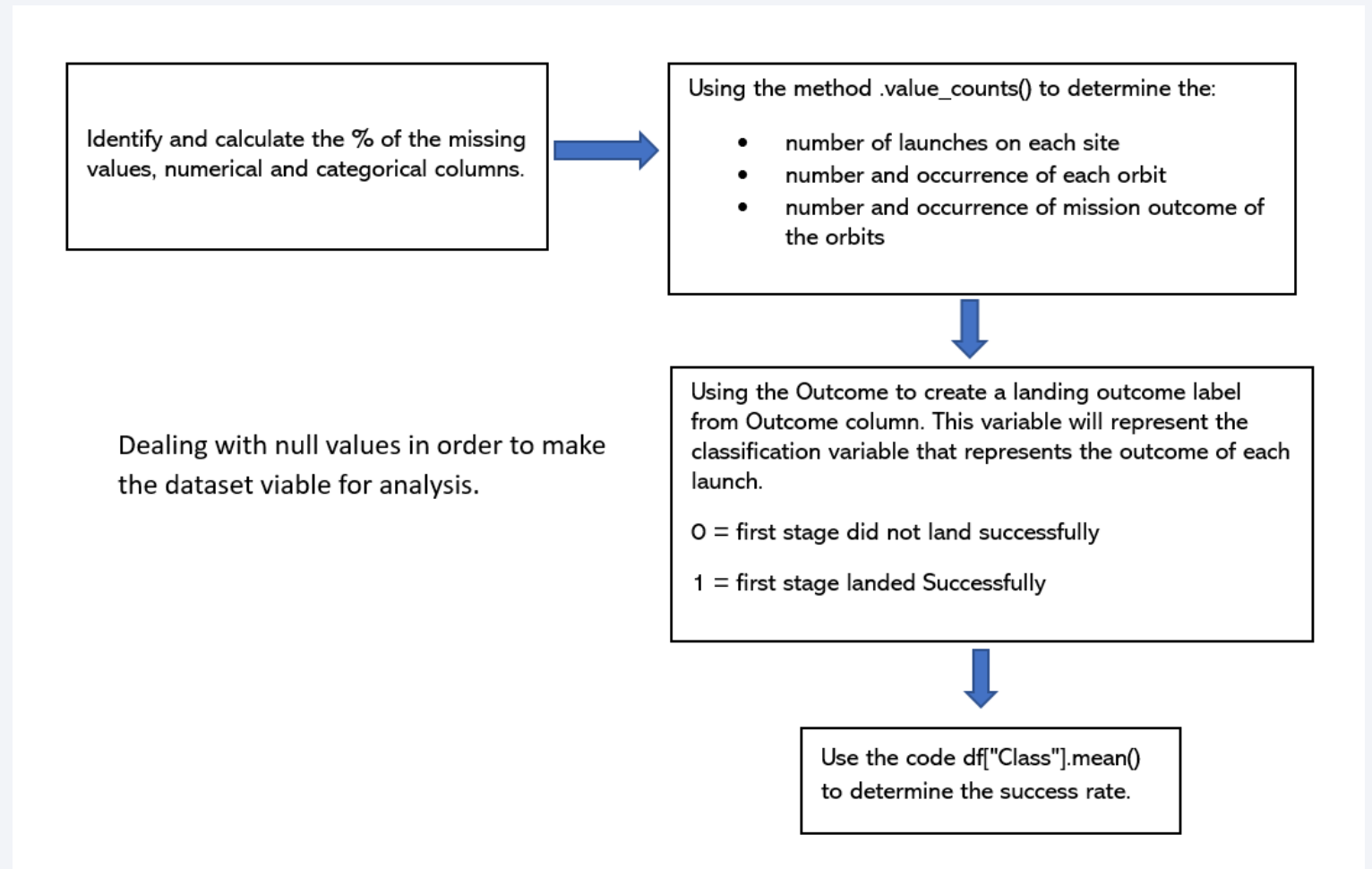


Data Wrangling

Exploratory data analysis and determining the training labels are performed.

GitHub URL of completed data wrangling related notebooks:

https://github.com/granaff/gafftestrepo/blob/1757cc84a14677a83fb97cc275c9642912951838/Data%20Wrangling_EDA.ipynb



EDA with Data Visualization

Summarize of charts plotted:

Scatter charts (Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload vs. Launch Site, Relationship between Flight Number and Orbit type, Relationship between Payload and Orbit type); **Bar chart** (Success rate of each orbit type); **Line chart** (Launch Success Yearly Trend)

Why you used those charts: To determine what attributes are correlated with successful landings.

GitHub URL of completed EDA with data visualization notebook:

<https://github.com/granaff/gafftestrepo/blob/1757cc84a14677a83fb97cc275c9642912951838/EDA%20with%20Visualization.ipynb>

EDA with SQL

SQL queries performed:

- Displaying the names of the unique launch sites in the space mission
- Displaying 5 records where launch sites begin with the string 'CCA'
- Displaying the total payload mass carried by boosters launched by NASA (CRS)
- Displaying average payload mass carried by booster version F9 v1.1
- Listing the date when the first successful landing outcome in ground pad was achieved.
- Listing the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- Listing the total number of successful and failure mission outcomes
- Listing the names of the booster_versions which have carried the maximum payload mass.
- Listing the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- Ranking the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

GitHub URL of completed EDA with SQL notebook:

<https://github.com/granaff/gafftestrepo/blob/ad1821e5066819046495802fb71bb5f1b4e41fef/EDA%20with%20SQL.ipynb>

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map

Map objects such as markers, circle, and lines are added to a folium map to help mark the launch site locations and their close proximities (to coastlines, to city, to railroad, to highway) on an interactive map. By doing so, we can explore the map with those markers to analyze and choose an optimal launch site.

- Explain why you added those objects: The launch success rate may also depend on the location and proximities of a launch site, i.e., the initial position of rocket trajectories.

GitHub URL of completed interactive map with Folium map:

<https://github.com/granaff/gafftestrepo/blob/bfe8c97a2ed3b53b39f6d77122c1b553b11b5adc/Interactive%20Visual%20Analytics%20with%20Folium.ipynb>

Build a Dashboard with Plotly Dash

An interactive dashboard with Plotly dash is built.

Pie charts are plotted showing the total launches by sites.

Scatter graph is plotted to show the relationship with Outcome and Payload Mass(kg) for different booster version

GitHub URL of completed Plotly Dash lab:

https://github.com/granaff/gafftestrepo/blob/1757cc84a14677a83fb97cc275c9642912951838/spacex_dash_app.py

Predictive Analysis (Classification)

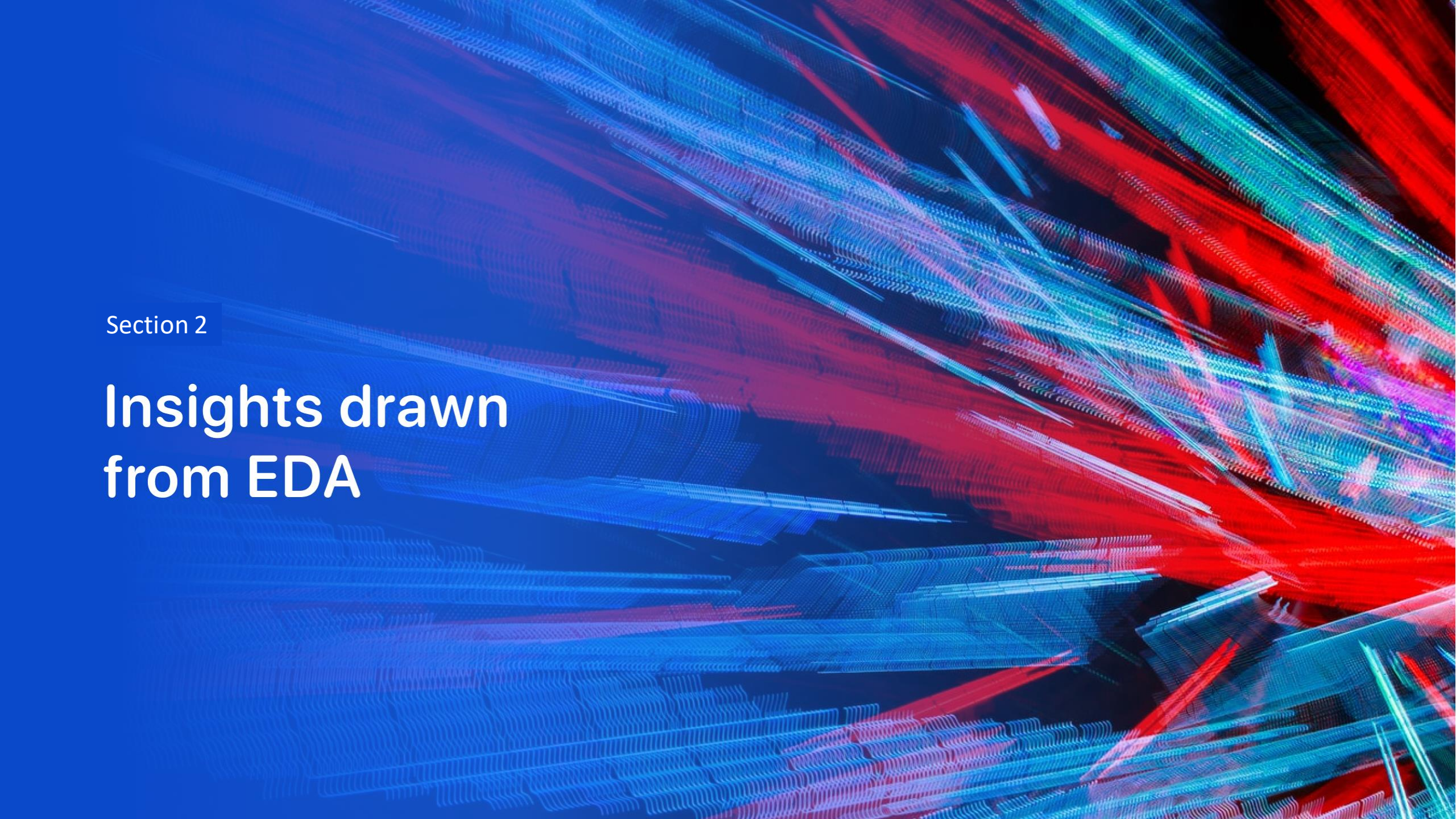
- The data are loaded using numpy and pandas, data are transformed and split into training and testing.
- Different machine learning models (Logistic Regression, SVM, KNN, DT) are built and tune different hyperparameters using GridSearch CV.
- Improved the model using feature engineering and algorithm tuning.
- Accuracy is used as the metric for the model; confusion matrices are built, too.

GitHub URL of completed predictive analysis lab:

<https://github.com/granaff/gafftestrepo/blob/f9e9d21801bd8a5fed0e0d92b00c2a14db22a9df/Machine%20Learning%20Prediction.ipynb>

Results

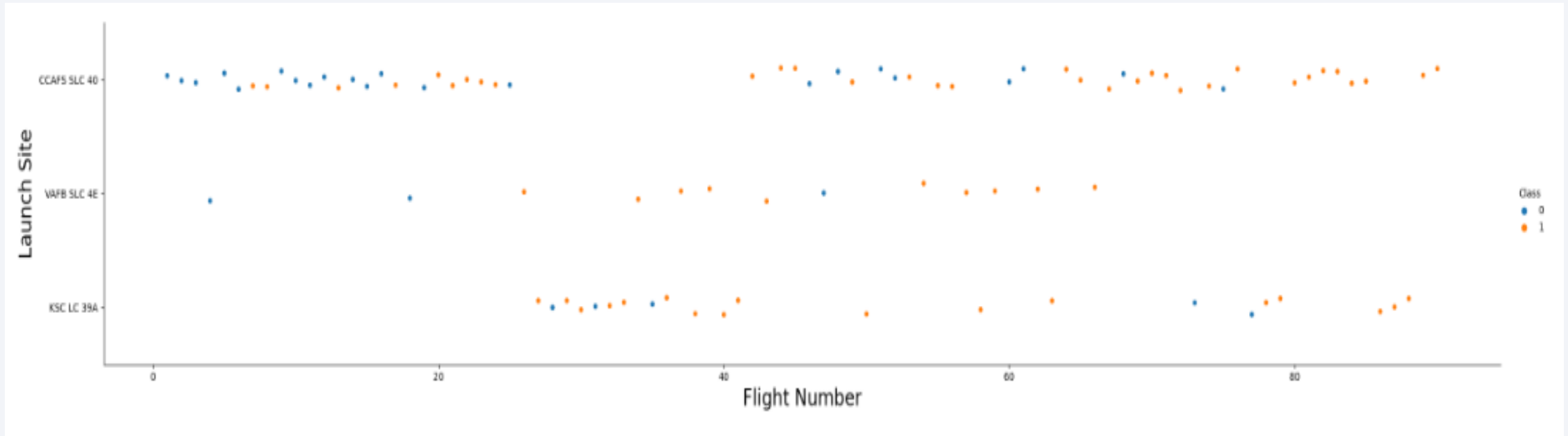
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

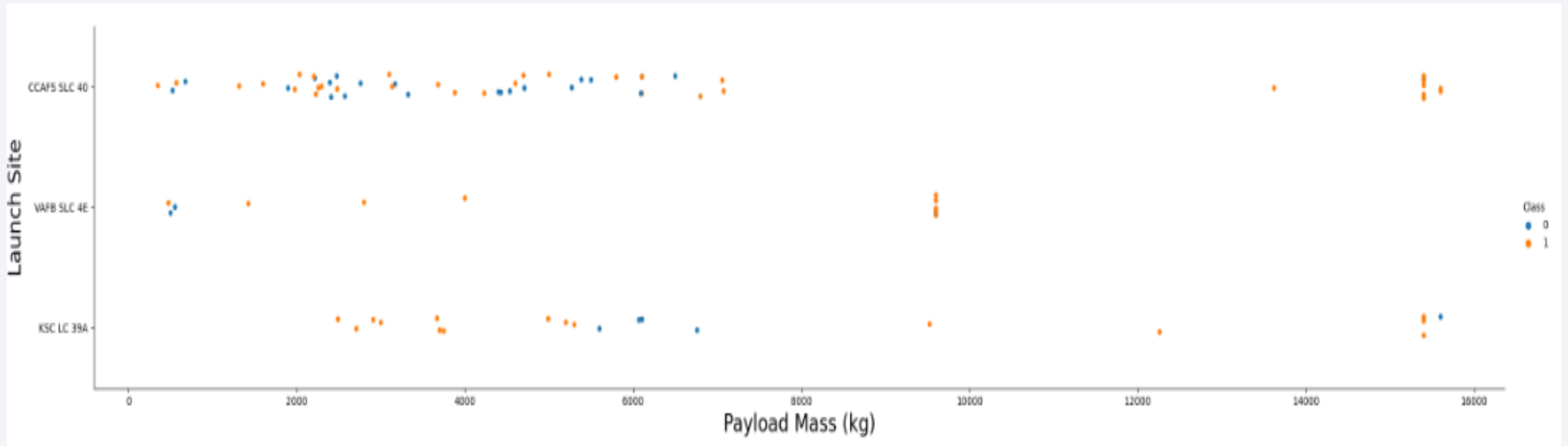
Insights drawn from EDA

Flight Number vs. Launch Site



The larger the flight number at a launch site, the greater the success rate at a launch site. Launches from the site of CCAFS SLC 40 are significantly higher than launches from other sites.

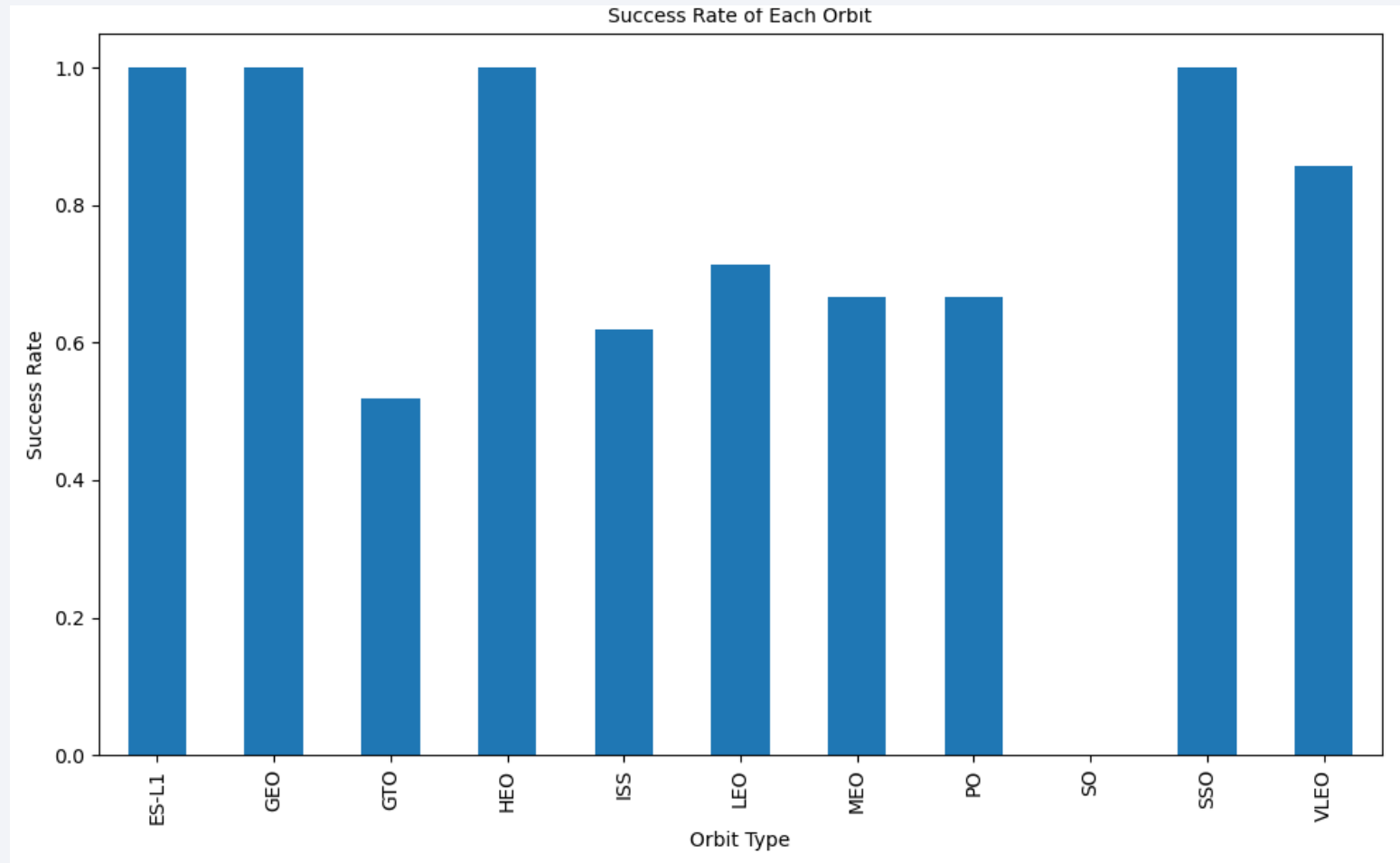
Payload vs. Launch Site



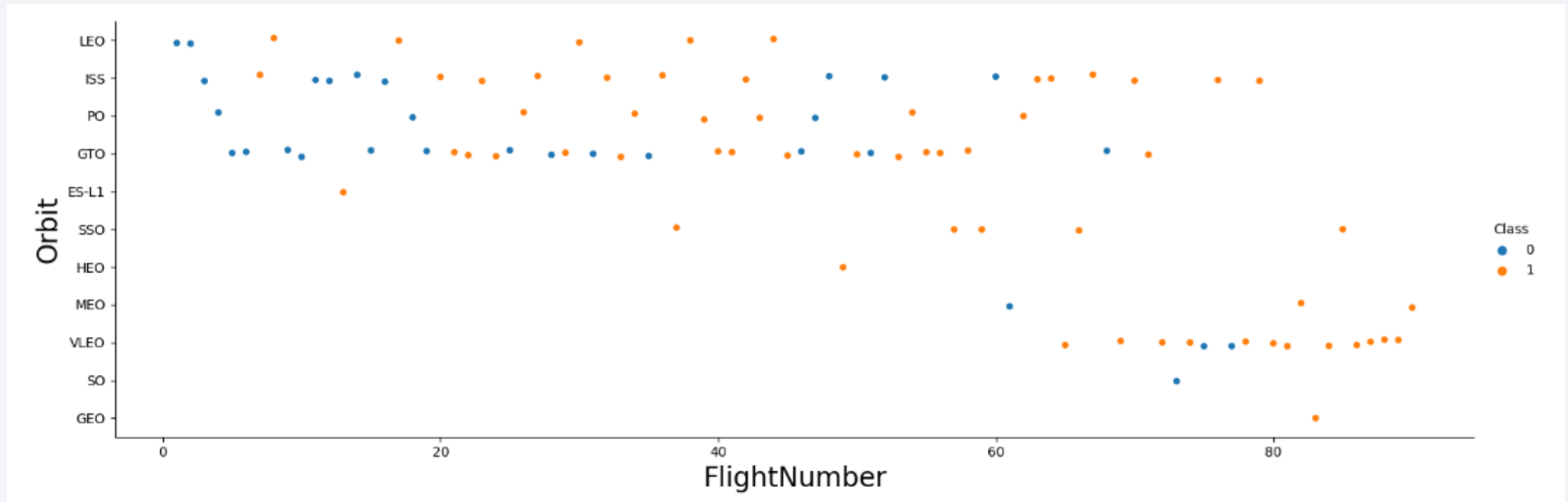
The majority of Payload with lower mass (kg) have been launched from CCAFS SLC 40. For VAFB-SLC launchsite there are no rockets launched for heavy payload mass(greater than 10000).

Success Rate vs. Orbit Type

Orbit types ES-L1, GEO, HEO, and SSO are among with the highest success rate.

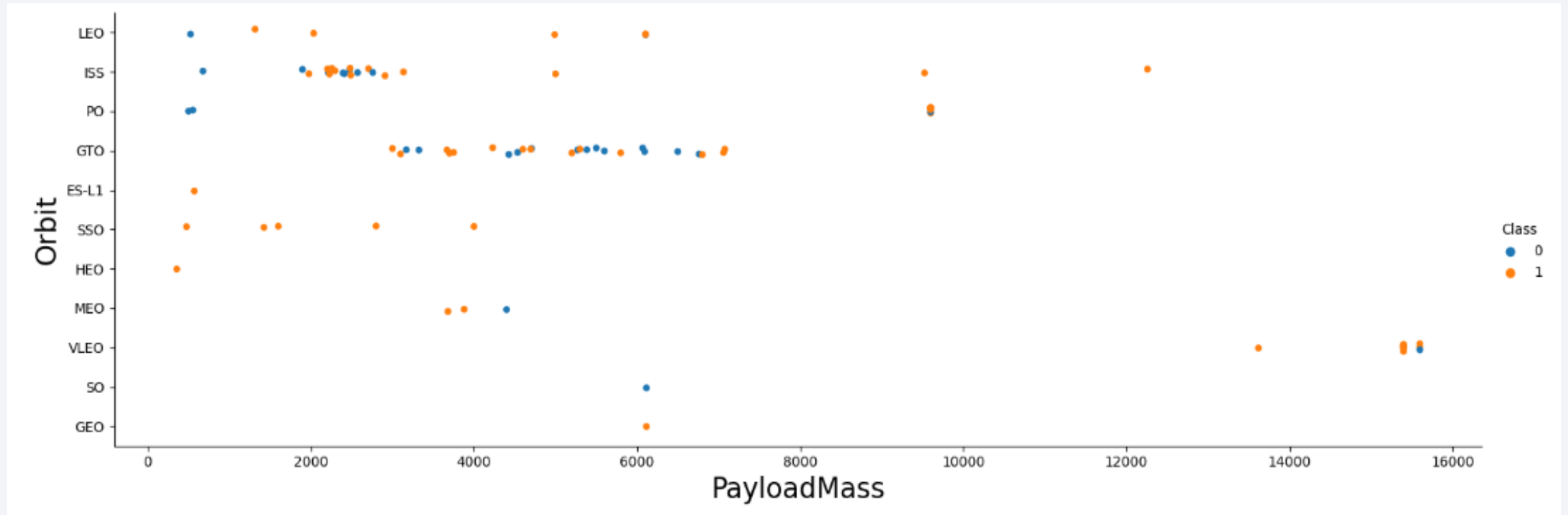


Flight Number vs. Orbit Type



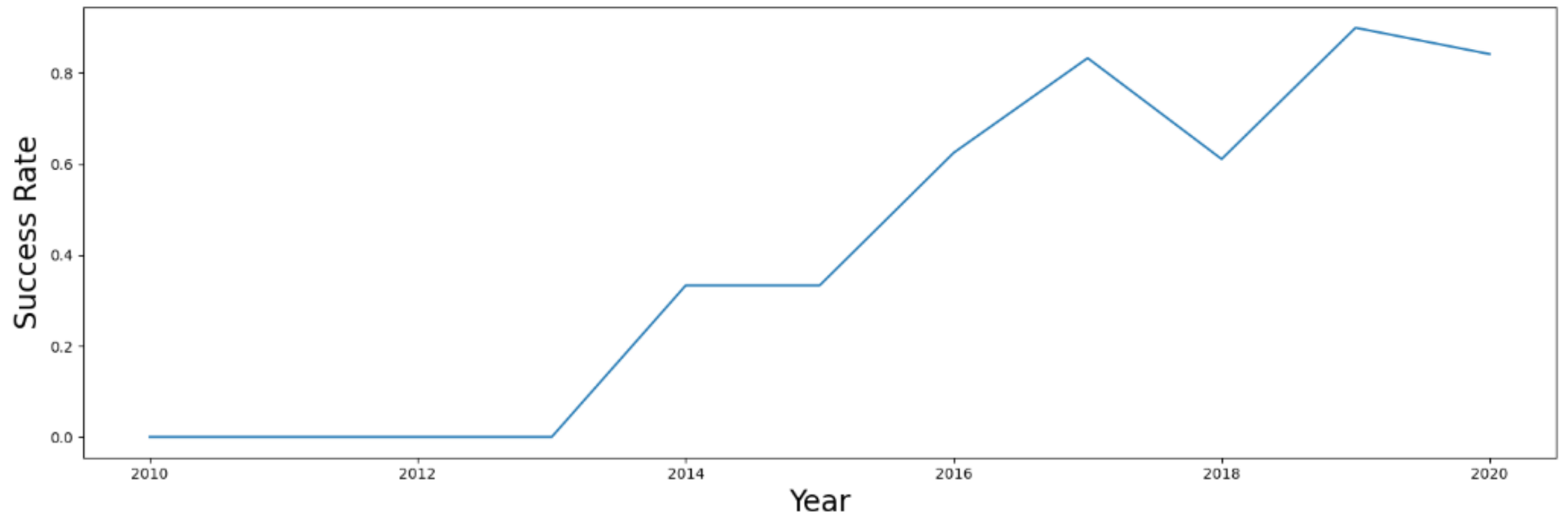
In the LEO orbit the success rate appears to be related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS. There are strong correlation between ISS and Payload at the range around 2000, as well as between GTO within the range of 4000-8000.

Launch Success Yearly Trend



The success rate since 2013 kept increasing till 2020.

All Launch Site Names

Display the names of the unique launch sites in the space mission

```
%sql select distinct(LAUNCH_SITE) from SPACEX
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

launch_site

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

DISTINCT key word is used to show only unique launch sites from the SpaceX data.

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%sql select * from SPACEX where LAUNCH_SITE like 'CCA%' limit 5
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB
Done.
```

DATE	time_utc	booster_version	launch_site	payload	payload_mass_kg	orbit	customer	mission_outcome	landing_outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

To display 5 records with launch sites begin with 'CCA', 'like 'CCA%' is added to the query. All the 5 records are referring to launch site CCAFS LC-40.

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql select sum(PAYLOAD_MASS_KG_) from SPACEX where CUSTOMER = 'NASA (CRS)'
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

1

45596

The total payload mass carried by boosters launched by NASA (CRS) is 45596.

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
%sql select avg(PAYLOAD_MASS_KG_) from SPACEX where BOOSTER_VERSION = 'F9 v1.1'
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

```
1
```

```
2928
```

The average payload mass by F9 v1.1 is 2928.

First Successful Ground Landing Date

List the date when the first successful landing outcome in ground pad was achieved.

Hint: Use min function

```
%sql select min(DATE) from SPACEX where Landing_Outcome = 'Success (ground pad)'
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

```
1
```

```
2015-12-22
```

On 2015-12-22, the first successful landing outcome in ground pad was achieved.

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql select BOOSTER_VERSION from SPACEX where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS_KG_ > 4000 and PAYLOAD_MASS_KG_ < 6000
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

booster_version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Booster version F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2 have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000.

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
%sql select mission_outcome, count(mission_outcome) from SPACEX GROUP BY mission_outcome
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

mission_outcome	2
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

The total number of successful (99) and failure (1) mission outcomes is 100.

Boosters Carried Maximum Payload

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
%sql select booster_version, payload_mass_kg_ from SPACEX where payload_mass_kg_ = (select max(payload_mass_kg_) from SPACEX)
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

booster_version	payload_mass_kg_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%sql select booster_version, launch_site from SPACEX where landing_outcome = 'Failure (drone ship)' and year(DATE) = 2015
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqn timer 39u98g.databases.appdomain.cloud:32286/BLUDB  
Done.
```

booster_version	launch_site
F9 v1.1 B1012	CCAFS LC-40
F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%sql select count(landing_outcome), landing_outcome from SPACEX \
where DATE between '2010-06-04' and '2017-03-20' group by landing_outcome\
order by count(landing_outcome) desc
```

```
* ibm_db_sa://kjm86934:***@1bbf73c5-d84a-4bb0-85b9-ab1a4348f4a4.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:32286/BLUDB
Done.
```

	landing_outcome
10	No attempt
5	Failure (drone ship)
5	Success (drone ship)
3	Controlled (ocean)
3	Success (ground pad)
2	Failure (parachute)
2	Uncontrolled (ocean)
1	Precluded (drone ship)

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark blue, with numerous bright yellow and orange lights representing cities and urban areas. The horizon line of the Earth is visible, separating the dark surface from the blackness of space.

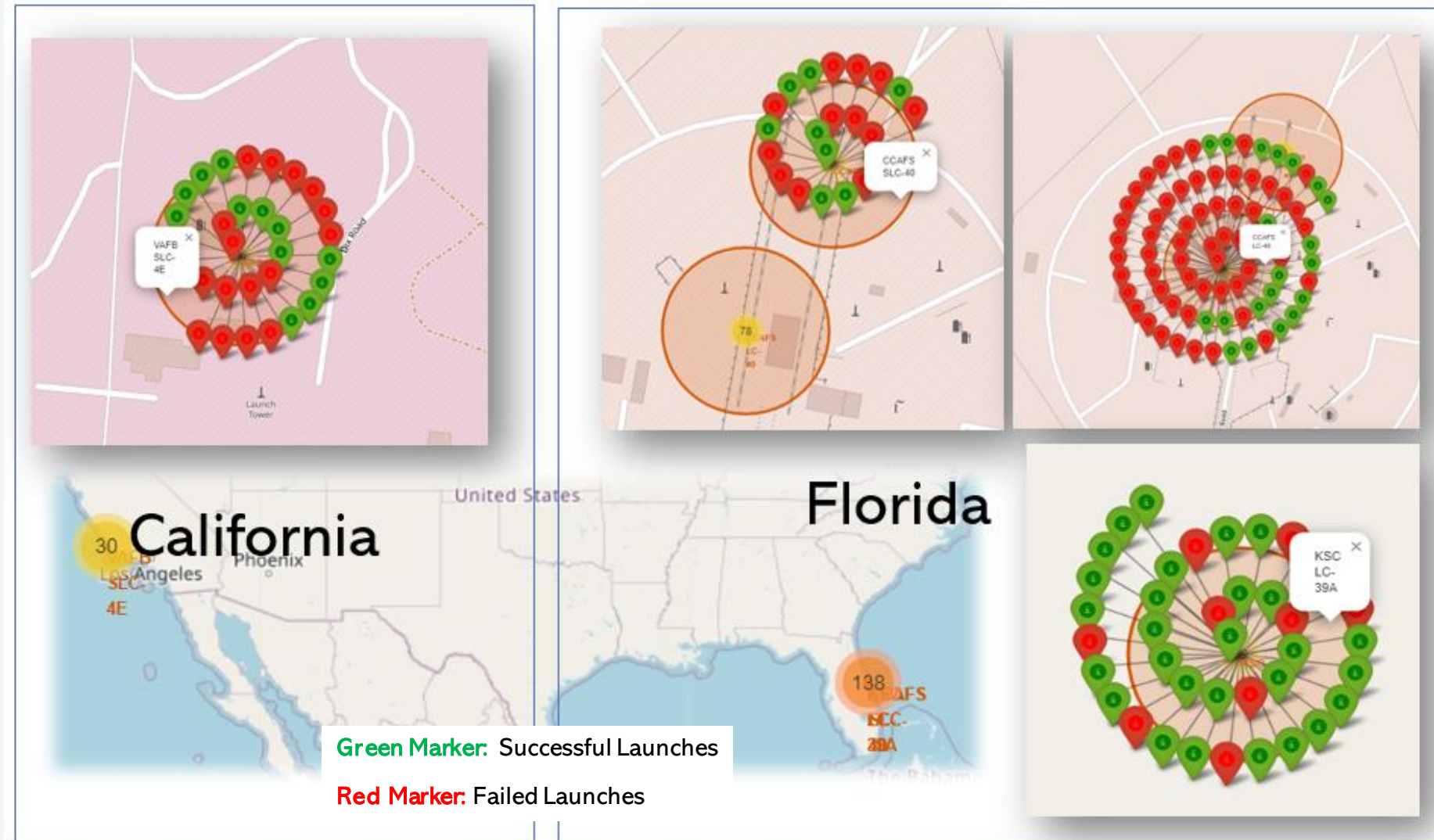
Section 3

Launch Sites Proximities Analysis

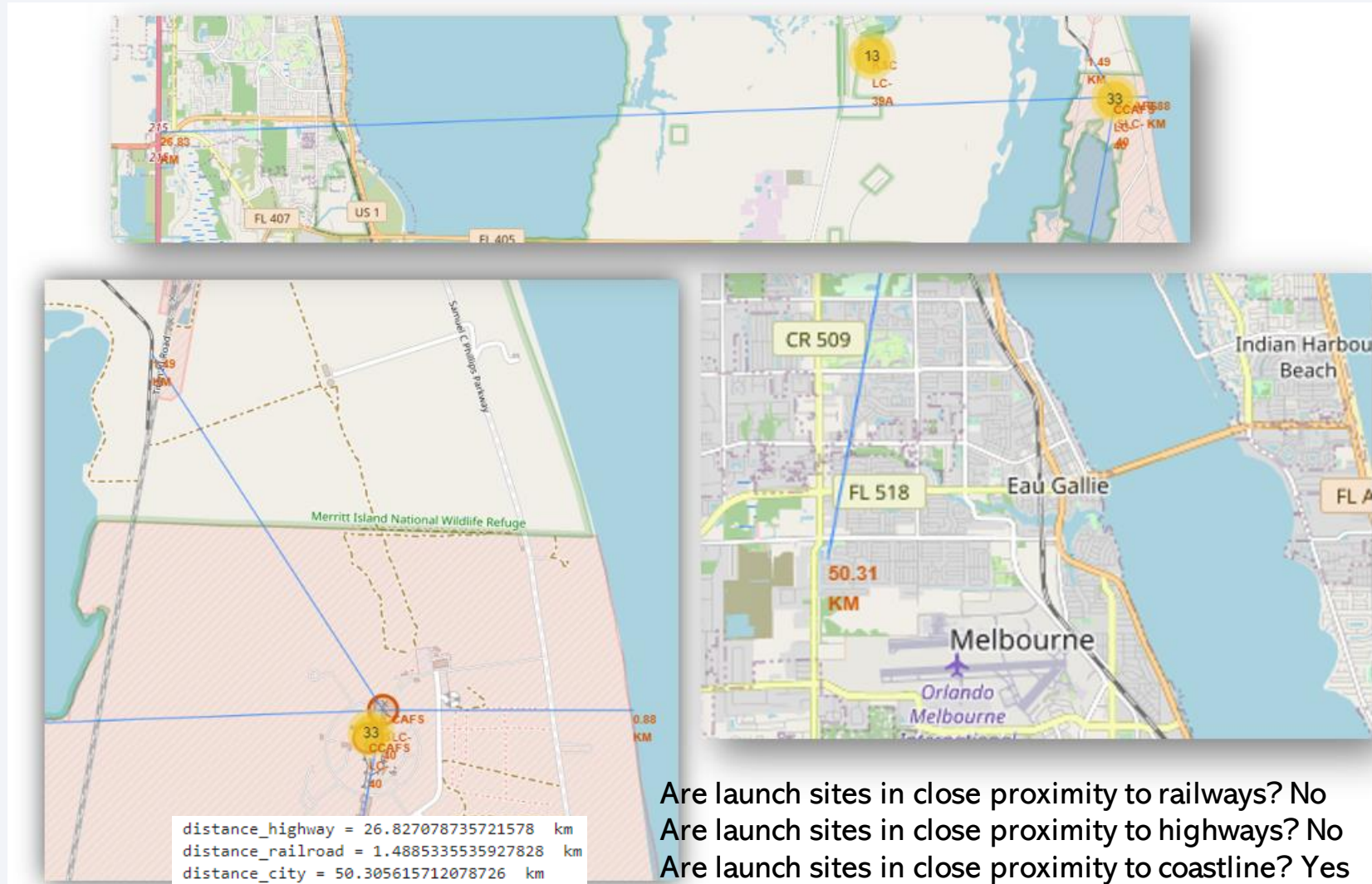
Launch Sites Location Markers on a Global Map



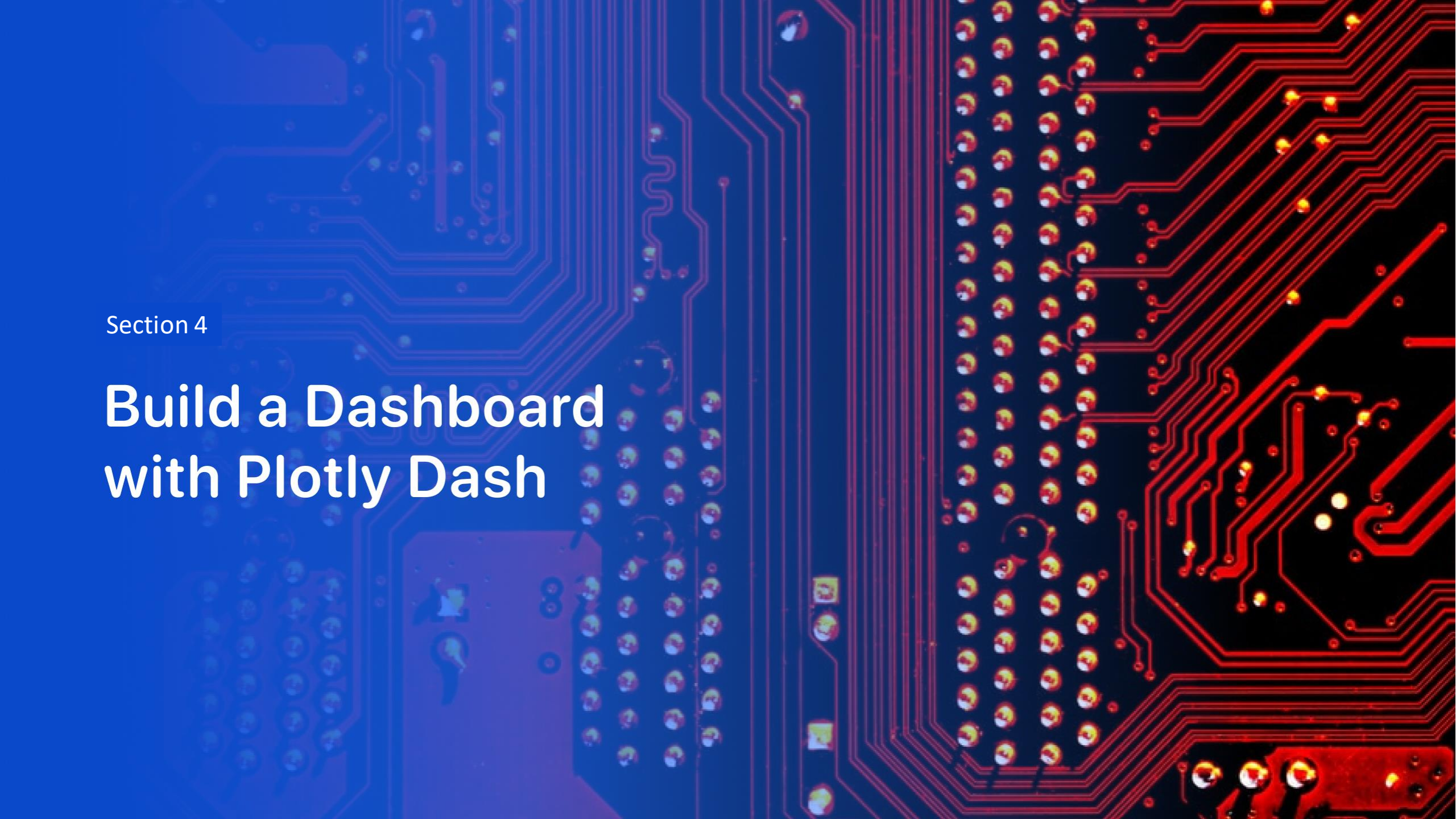
Launch Outcomes on the Map



Launch site to its Proximities



Are launch sites in close proximity to railways? No
Are launch sites in close proximity to highways? No
Are launch sites in close proximity to coastline? Yes
Do launch sites keep certain distance away from cities? No



Section 4

Build a Dashboard with Plotly Dash

Launch Success Percentage for All Sites

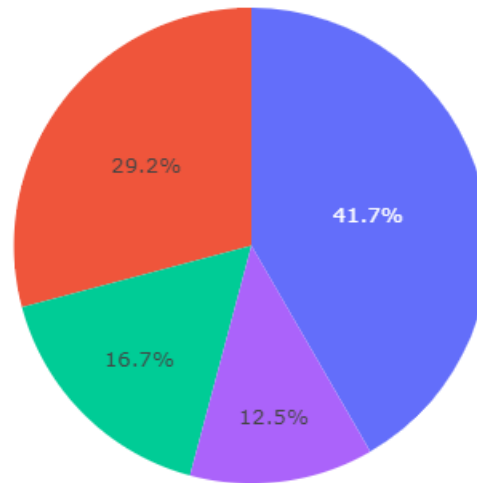
SpaceX Launch Records Dashboard

All Sites



Success Count for all launch sites

KSC LC-39A has the most successful launches among all sites.



■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40

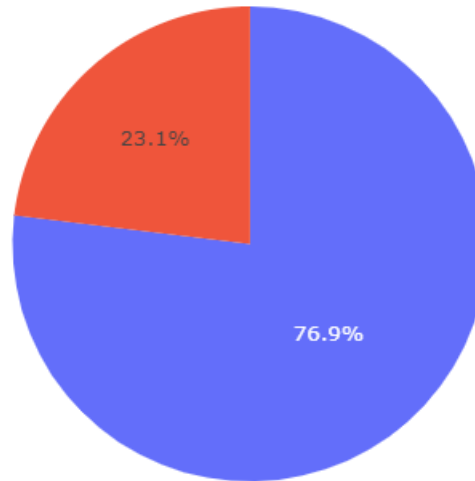
Launch site with highest launch success ratio

SpaceX Launch Records Dashboard

Kennedy Space Center Launch Complex 39A (KSC LC-39A) × ▼

Total Success Launches for site KSC LC-39A

KSC LC-39A has
success rate of 76.9%
and a failure rate of
23.1%.



Launch Outcome scatter plot for all sites

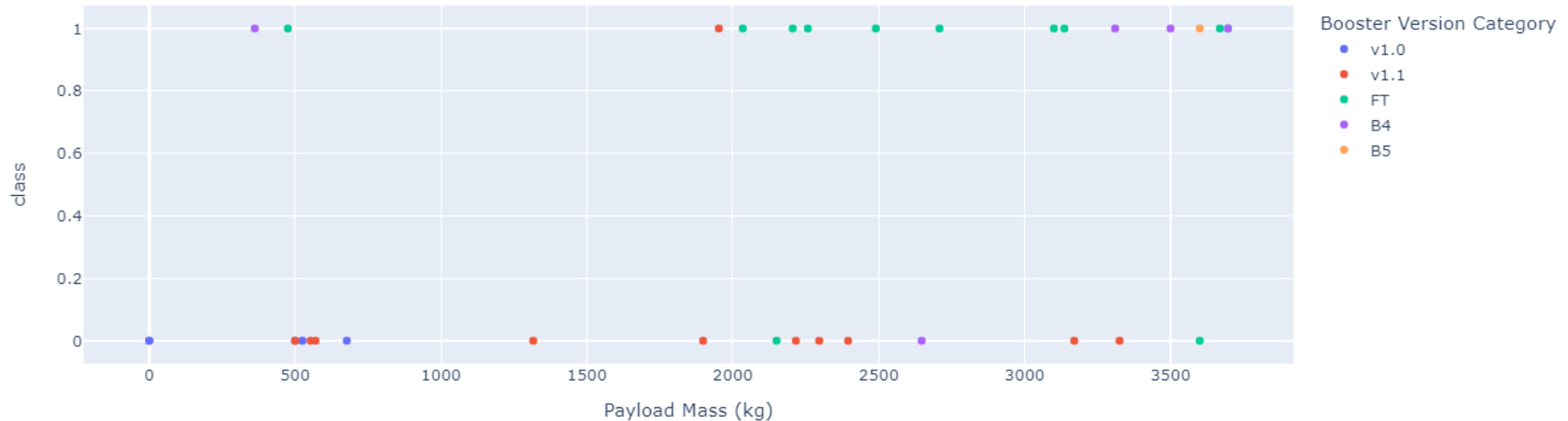
Low Weighted Payload 0kg-4000kg

Payload range (Kg):



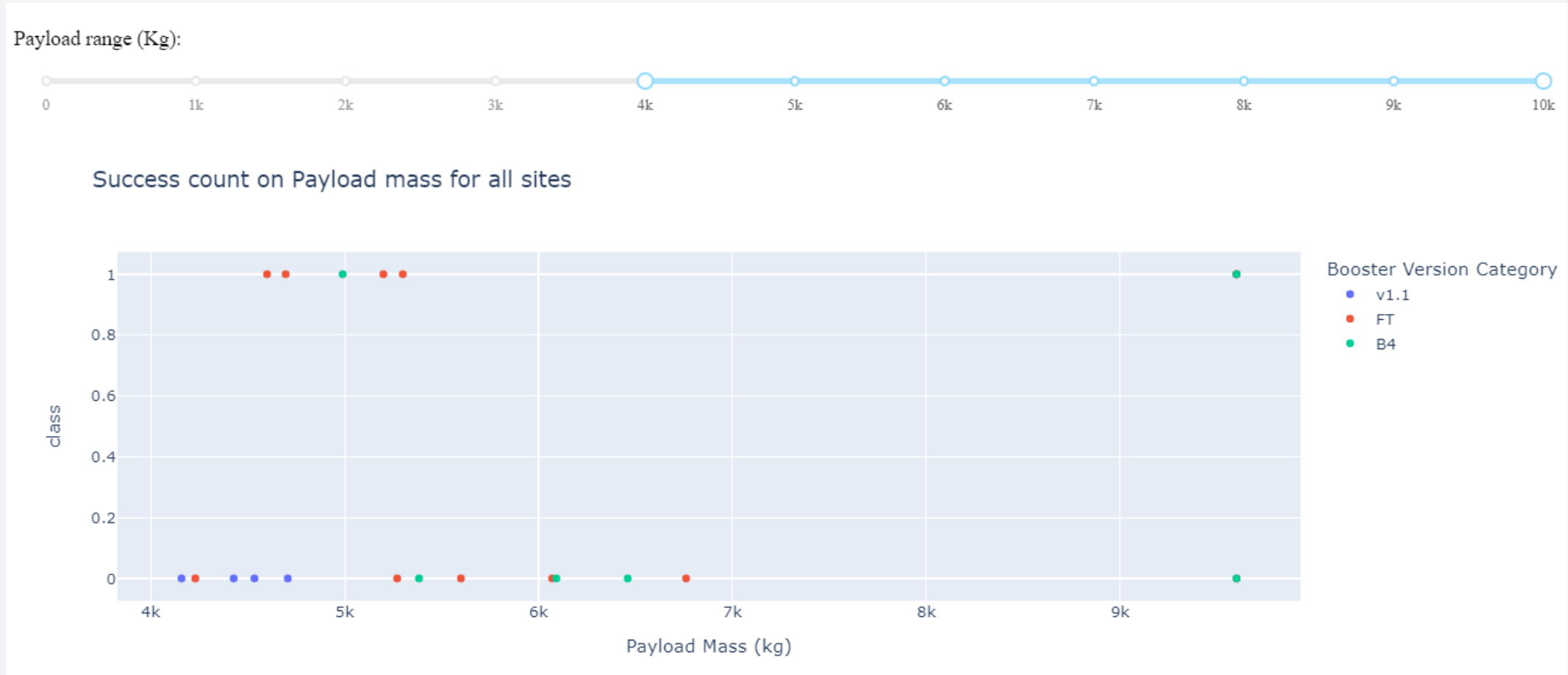
Success rates for low weighted payloads are higher than heavy weighted payloads.

Success count on Payload mass for all sites



Launch Outcome scatter plot for all sites

Heavy Weighted Payload 4000kg-10000kg

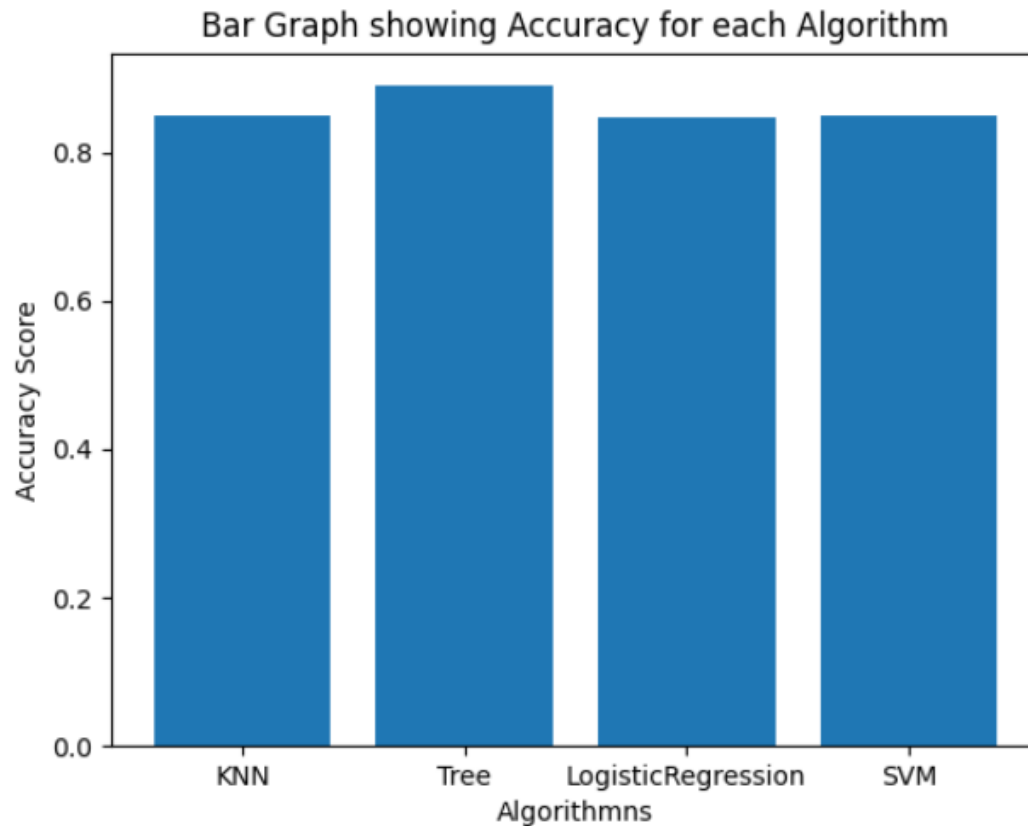




Section 5

Predictive Analysis (Classification)

Classification Accuracy



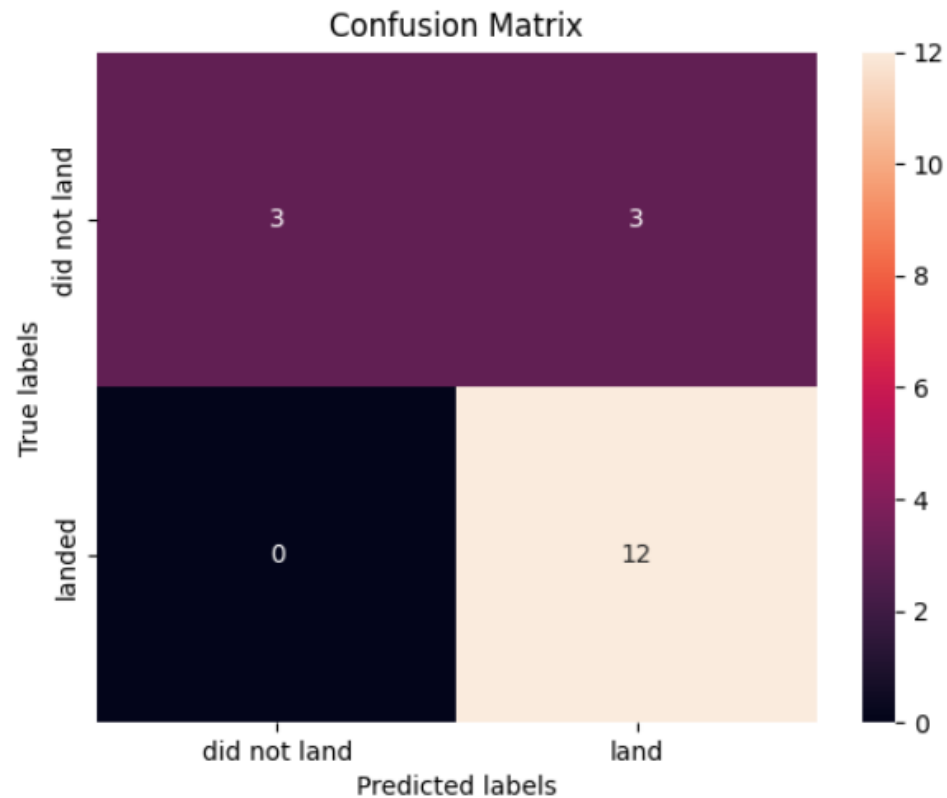
The decision tree classifier is the model with the highest classification accuracy

Best Algorithm is Tree with a score of 0.8892857142857145
Best Params is : {'criterion': 'entropy', 'max_depth': 8, 'max_features': 'auto', 'min_samples_leaf': 2, 'min_samples_split': 10, 'splitter': 'random'}

Confusion Matrix

We can plot the confusion matrix

```
: yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



FN (False Positives)
Unsuccessful landing marked as
successful landing by the classifier.

Conclusions

- As the flight number increases, the first stage is more likely to land successfully.
- The more massive the payload, the less likely the first stage will return.
- Launch success rate is increasing since 2013 until 2020.
- Orbits ES-L1, GEO, HEO, SSO, VLEO have the most success rates.
- KSC LC-39A had the most successful launches of any sites.
- The decision tree classifier is the model with the highest classification accuracy for this task.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Github link: <https://github.com/granaff/gafftestrepo.git>

Thank you!

